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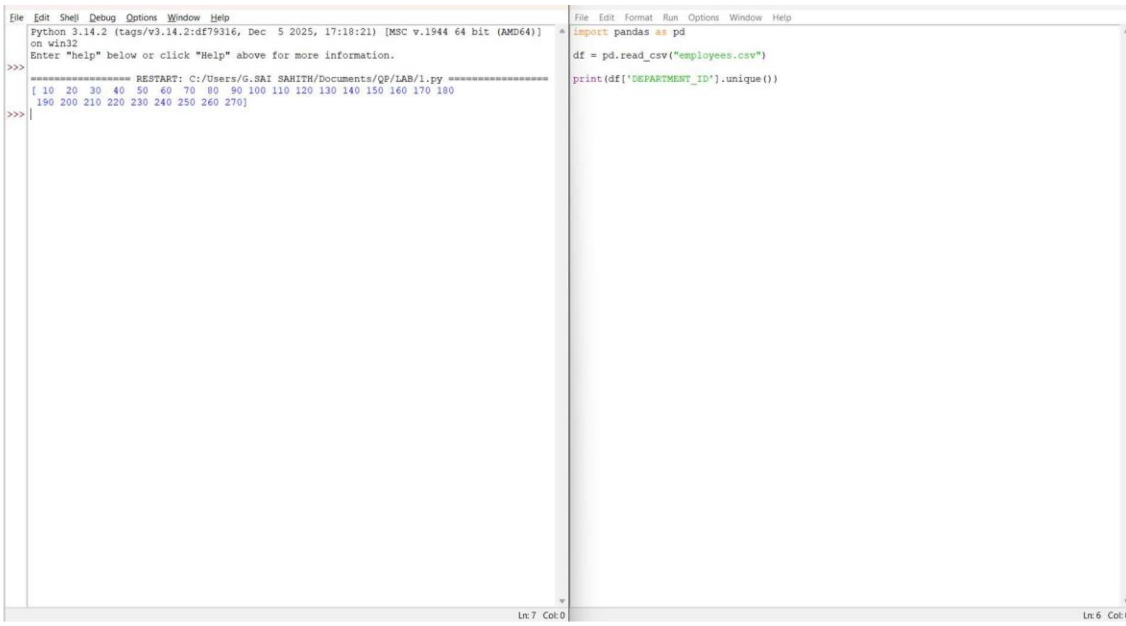
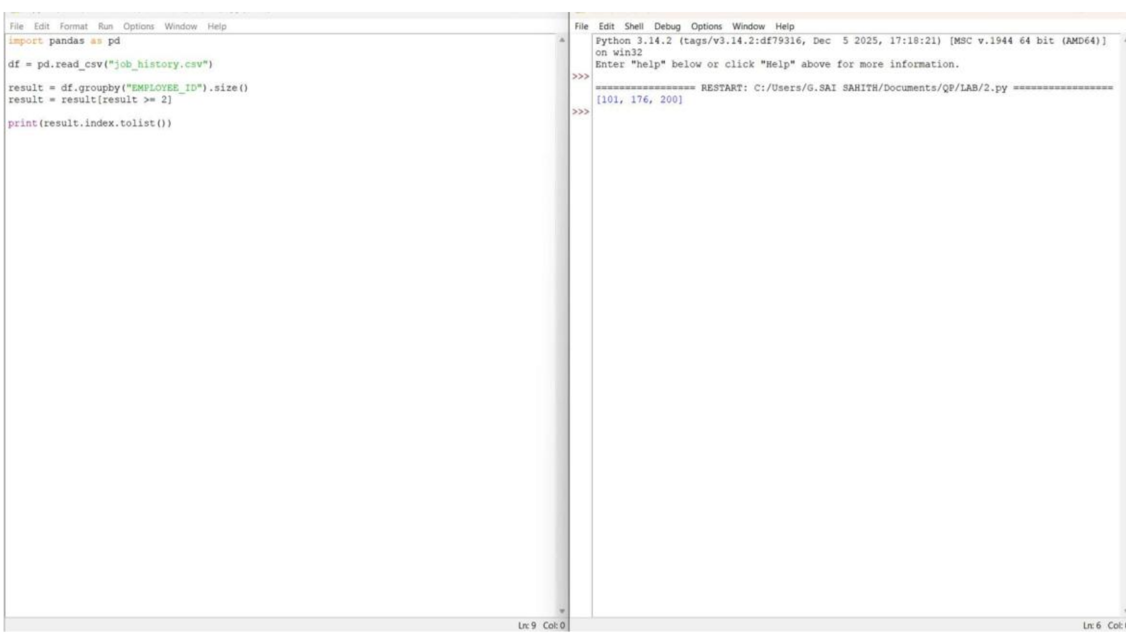
REG NO	192424213
COURSE CODE	DSA 0515
COURSE NAME	QUERY PROCESSING FOR DATA SCIENCE IN SQL ENGINES
LAB OUTPUTS	1 - 20

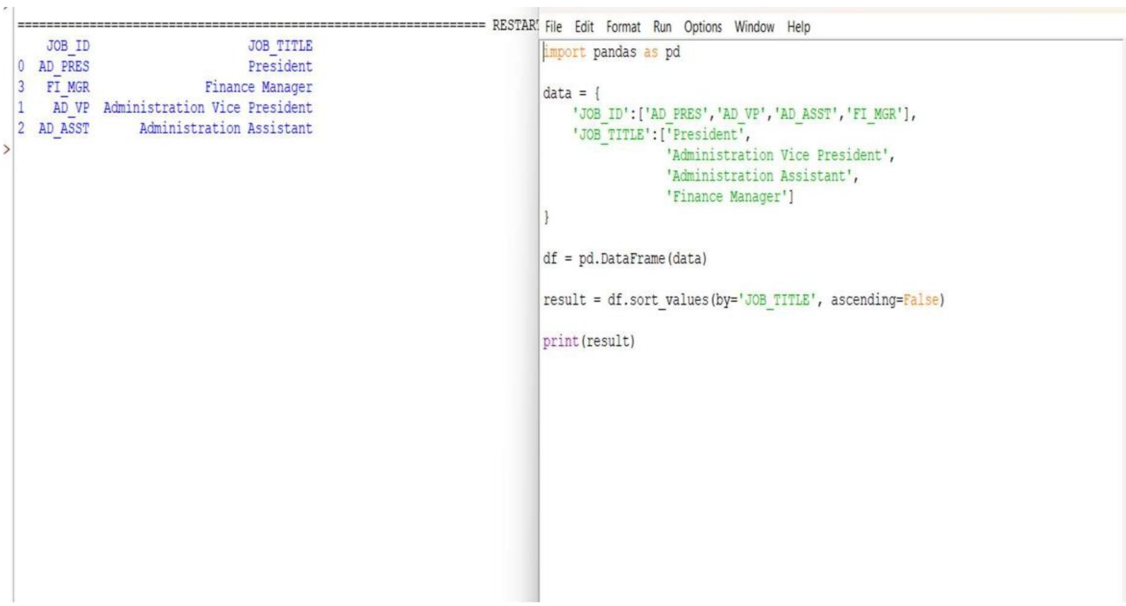
EXP-1


File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/1.py ================
[10 20 30 40 50 60 70 80 90 100 110 120 130 140 150 160 170 180
190 200 210 220 230 240 250 260 270]
>>>
Ln: 7 Col: 0
File Edit Format Run Options Window Help
import pandas as pd
df = pd.read_csv("employees.csv")
print(df['DEPARTMENT_ID'].unique())
Ln: 6 Col: 0

EXP-2

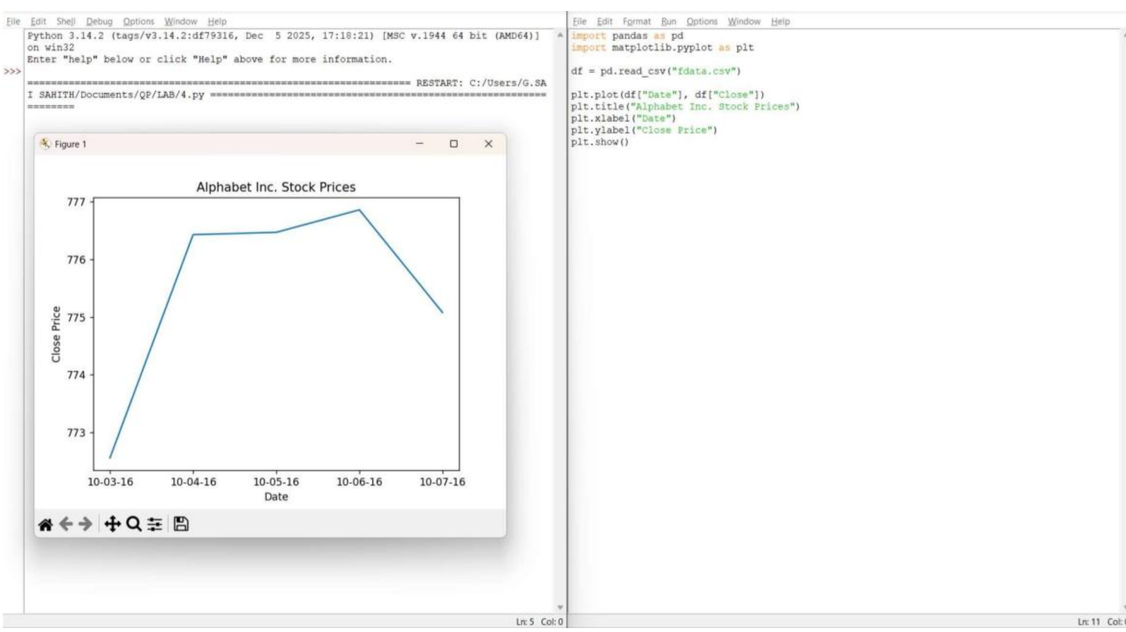

File Edit Format Run Options Window Help
import pandas as pd
df = pd.read_csv("job_history.csv")
result = df.groupby("EMPLOYEE_ID").size()
result = result[result >= 2]
print(result.index.tolist())
Ln: 9 Col: 0
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/2.py ================
[101, 176, 200]
>>>
Ln: 6 Col: 0

EXP-3



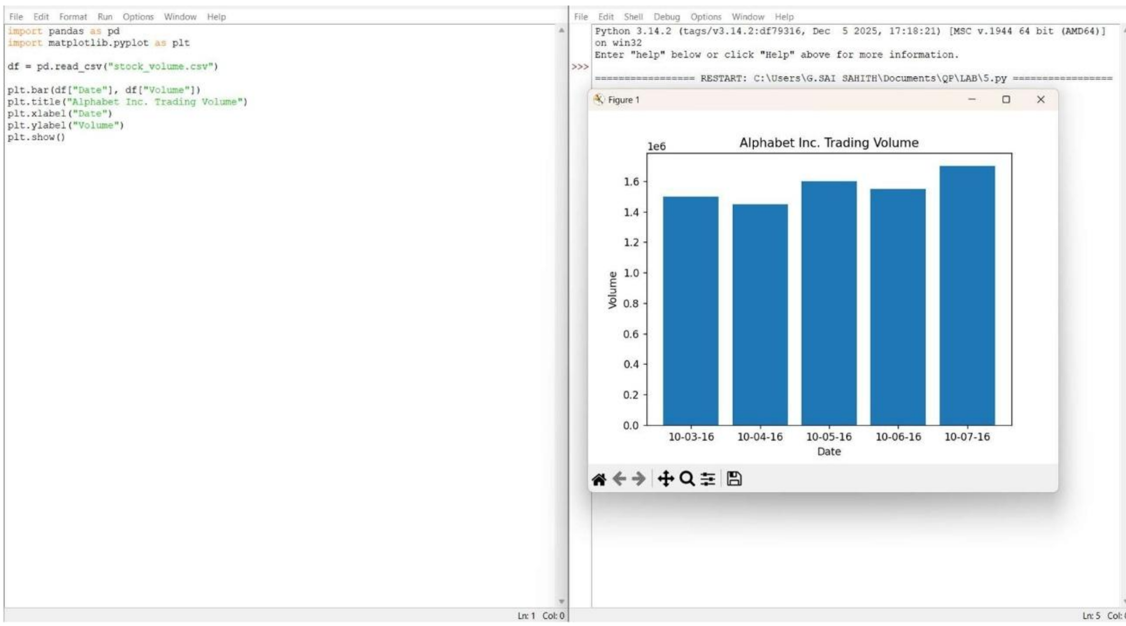
JOB_ID JOB_TITLE
0 AD_PRES President
3 FI_MGR Finance Manager
1 AD_VP Administration Vice President
2 AD_ASST Administration Assistant
File Edit Format Run Options Window Help
import pandas as pd
data = {
'JOB_ID':['AD_PRES','AD_VP','AD_ASST','FI_MGR'],
'JOB_TITLE':['President',
'Administration Vice President',
'Administration Assistant',
'Finance Manager']
}
df = pd.DataFrame(data)
result = df.sort_values(by='JOB_TITLE', ascending=False)
print(result)

EXP-4

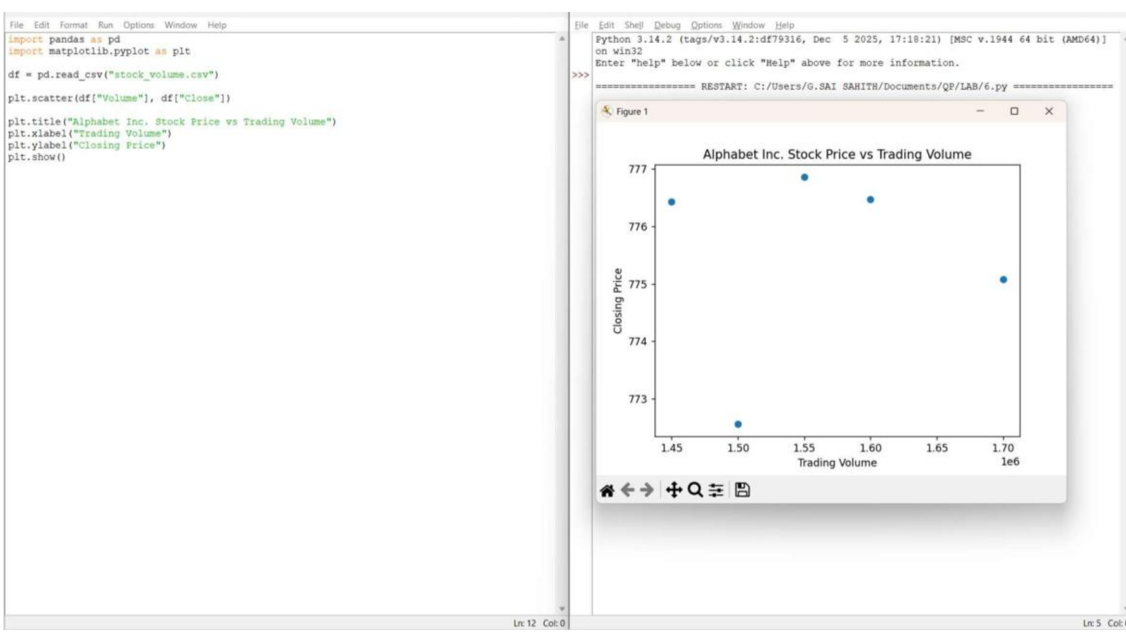


File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/4.py
Figure 1
Alphabet Inc. Stock Prices
Close Price
Date
10-03-16
10-04-16
10-05-16
10-06-16
10-07-16
File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("fdata.csv")
plt.plot(df["Date"], df["Close"])
plt.title("Alphabet Inc. Stock Prices")
plt.xlabel("Date")
plt.ylabel("Close Price")
plt.show()

EXP-5


File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("stock_volume.csv")
plt.bar(df["Date"], df["Volume"])
plt.title("Alphabet Inc. Trading Volume")
plt.xlabel("Date")
plt.ylabel("Volume")
plt.show()
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:\Users\G.SAI SAHITH\Documents\QP\LAB\5.py ================
Figure 1
Alphabet Inc. Trading Volume
1e6
Volume
Date
10-03-16
10-04-16
10-05-16
10-06-16
10-07-16
Ln: 1 Col: 0
Ln: 5 Col: 0

EXP-6


File Edit Format Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv("stock_volume.csv")
plt.scatter(df["Volume"], df["Close"])
plt.title("Alphabet Inc. Stock Price vs Trading Volume")
plt.xlabel("Trading Volume")
plt.ylabel("Closing Price")
plt.show()
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/6.py ================
Figure 1
Alphabet Inc. Stock Price vs Trading Volume
Closing Price
Trading Volume
1e6
Ln: 12 Col: 0
Ln: 5 Col: 0

EXP-7

File Edit Format Run Options Window Help

```python
import pandas as pd

df = pd.read_csv("sales_data.csv")

pivot = pd.pivot_table(
    df,
    values="Sale_Value",
    index="Item",
    aggfunc=["max", "min"]
)

print(pivot)
```

Ln: 13 Col: 0

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/7.py =================
              max        min
       Sale_Value Sale_Value
Item
Laptop      50000      45000
Mobile      25000      20000
Tablet      18000      15000
>>>
```

Ln: 11 Col: 0

EXP-8

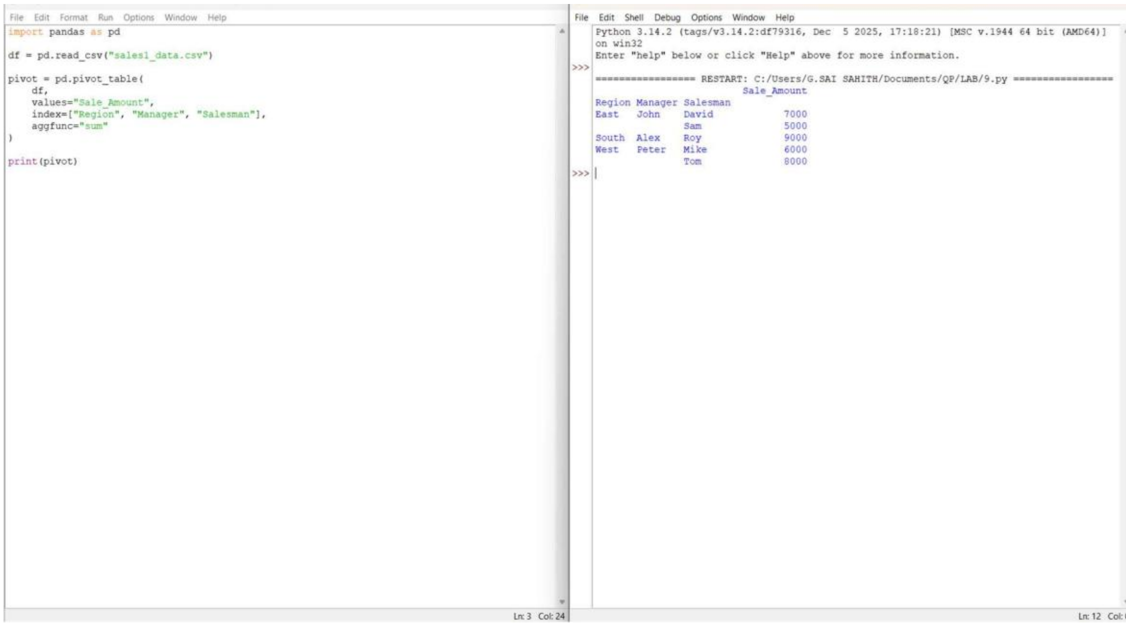
File Edit Format Run Options Window Help

```python
import pandas as pd

df = pd.read_csv("sales_units.csv")

pivot = pd.pivot_table(
    df,
    values='Units_Sold',
    index='Item',
    aggfunc='sum'
)

print(pivot)
```

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: C:\Users\G.SAI SAHITH\Documents\QP\LAB\8.py =================
        Units_Sold
Item
Laptop          22
Mobile          38
Tablet          25
>>>
```

EXP-9



File Edit Format Run Options Window Help

```
import pandas as pd

df = pd.read_csv("sales1_data.csv")

pivot = pd.pivot_table(
    df,
    values="Sale_Amount",
    index=["Region", "Manager", "Salesman"],
    aggfunc="sum"
)

print(pivot)
```

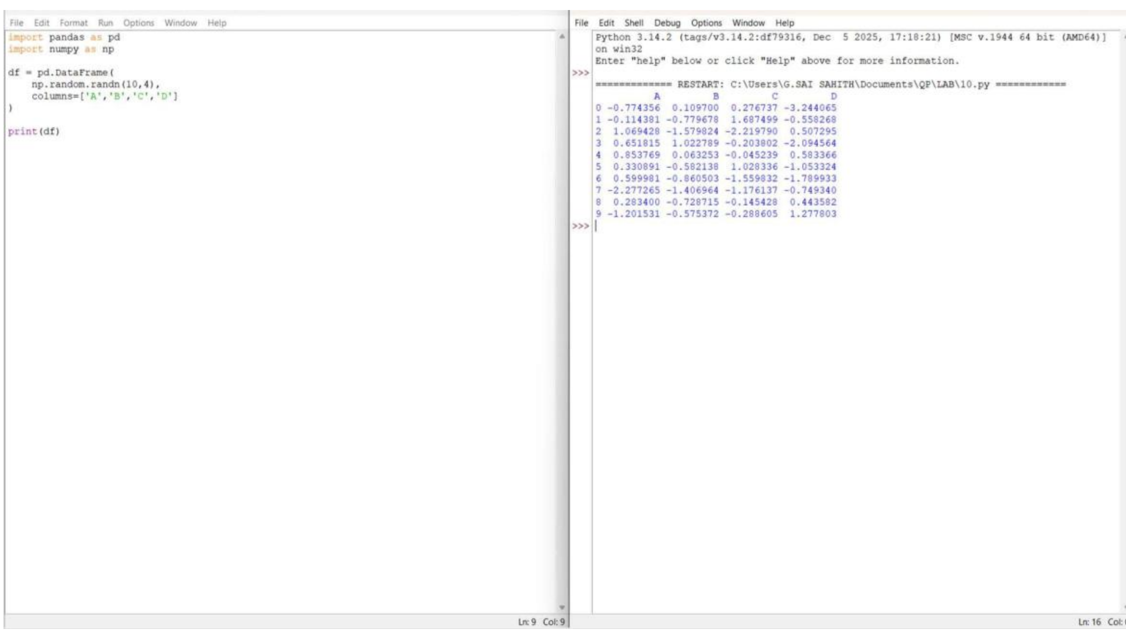
Ln: 3 Col: 24

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/9.py =================
                         Sale_Amount
Region Manager Salesman
East   John    David            7000
               Sam              5000
South  Alex    Roy              9000
West   Peter   Mike             6000
               Tom              8000
>>>
```

Ln: 12 Col: 0

EXP-10



File Edit Format Run Options Window Help

```
import pandas as pd
import numpy as np

df = pd.DataFrame(
    np.random.randn(10,4),
    columns=['A','B','C','D']
)

print(df)
```

Ln: 9 Col: 9

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
============= RESTART: C:\Users\G.SAI SAHITH\Documents\QP\LAB\10.py ============
          A         B         C         D
0 -0.774356  0.109700  0.276737 -3.244065
1 -0.114381 -0.779678  1.687499 -0.558268
2  1.069428 -1.579824 -2.219790  0.507295
3  0.651815  1.022789 -0.203802 -2.094564
4  0.853769  0.063253 -0.045239  0.583366
5  0.330891 -0.582138  1.028336 -1.053324
6  0.599981 -0.860503 -1.559832 -1.789933
7 -2.277265 -1.406964 -1.176137 -0.749340
8  0.283400 -0.728715 -0.145428  0.443582
9 -1.201531 -0.575372 -0.288605  1.277803
>>>
```

Ln: 16 Col: 0

EXP-11

```
ile  Edit  Format  Run  Options  Window  Help
mport pandas as pd
mport numpy as np

 Create DataFrame
lf = pd.DataFrame(np.random.randn(10,4),
                  columns=['A','B','C','D'])

 Convert some values to NaN
lf.iloc[2,1] = np.nan
lf.iloc[5,3] = np.nan
lf.iloc[7,0] = np.nan

rint("DataFrame:")
rint(df)

rint("\nNaN Values:")
rint(df.isnull())
```

Ln: 18 Col: 0

```
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
================ RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/11.py ================
DataFrame:
          A         B         C         D
0  1.320193  0.664389 -1.412324 -1.145675
1 -0.714726 -1.089827 -0.372183 -0.144428
2  1.211815       NaN  1.434420 -0.345904
3  1.613146  1.291952 -0.410312  0.658473
4 -1.094787  0.815967  1.925427 -0.625744
5 -0.153027 -0.113928  0.118097       NaN
6 -0.587423 -0.456434  0.209854 -0.692433
7       NaN -1.127333  0.547612  0.976376
8  0.711657  2.487590  0.769902  1.293178
9  0.298399  0.652405  0.466473 -0.266635

NaN Values:
       A      B      C      D
0  False  False  False  False
1  False  False  False  False
2  False   True  False  False
3  False  False  False  False
4  False  False  False  False
5  False  False  False   True
6  False  False  False  False
7   True  False  False  False
8  False  False  False  False
9  False  False  False  False
>>>
```

Ln: 30 Col: 0

EXP-12

```python
import pandas as pd
import numpy as np

df = pd.DataFrame(np.random.randn(10,4),
                  columns=['A','B','C','D'])

df.style.set_properties(**{
    'background-color':'black',
    'color':'yellow'
})
```

	A	B	C	D
0	0.582516	0.860781	-0.364376	-1.671961
1	0.361479	0.179580	-3.352339	-0.153773
2	-1.034816	0.606783	0.384148	2.266582
3	-1.635183	-0.487048	-1.233669	1.197715
4	-1.263204	0.446833	-0.789460	-1.401966
5	-0.496572	-0.798043	-0.768905	-0.856221
6	0.296654	-1.577515	-0.772830	1.579341
7	0.836582	-0.768561	0.439611	-1.839904
8	-0.932523	0.111614	1.507532	0.688467
9	-1.480248	-0.057373	1.017798	0.498341

EXP-13

File Edit Format Run Options Window Help

```python
import pandas as pd
import numpy as np

df = pd.DataFrame({
    'A':[1,2,np.nan,4],
    'B':[5,np.nan,7,8],
    'C':[9,10,11,np.nan]
})

print(df.isnull())
```

Ln: 11 Col: 0

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/13.py ================
       A      B      C
0  False  False  False
1  False   True  False
2   True  False  False
3  False  False   True
>>>
```

Ln: 10 Col: 0

EXP-14

File Edit Format Run Options Window Help

```python
import pandas as pd
import numpy as np

df = pd.DataFrame({
    'A':[1,2,np.nan,4],
    'B':[5,np.nan,7,8],
    'C':[9,10,11,np.nan]
})

df.fillna(0, inplace=True)

print(df)
```

Ln: 13 Col: 0

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/14.py ================
     A    B     C
0  1.0  5.0   9.0
1  2.0  0.0  10.0
2  0.0  7.0  11.0
3  4.0  8.0   0.0
>>>
```

Ln: 10 Col: 0

EXP-15:

```python
import pandas as pd
import numpy as np

df = pd.DataFrame({
    'A':[1,np.nan,np.nan,4],
    'B':[np.nan,np.nan,7,8],
    'C':[9,np.nan,np.nan,np.nan]
})

result = df[df.isnull().sum(axis=1) >= 2]

print(result)
```

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
================= RESTART: C:/Users/G.SAI SAHITH/Documents/QP/LAB/15.py ================
     A    B   C
1 NaN  NaN NaN
2 NaN  7.0 NaN
>>>
```

EXP-16

```python
import pandas as pd

df = pd.DataFrame({
    'School_Code':['S001','S002','S001','S002','S003'],
    'Name':['A','B','C','D','E'],
    'Age':[15,16,15,17,16]
})

grouped = df.groupby('School_Code')

print(grouped)

for name, group in grouped:
    print("\nSchool Code:", name)
    print(group)

print("\nType of GroupBy Object:")
print(type(grouped))
```

```
<pandas.core.groupby.generic.DataFrameGroupBy object at 0x787fabd445f0>

School Code: S001
  School_Code Name  Age
0        S001    A   15
2        S001    C   15

School Code: S002
  School_Code Name  Age
1        S002    B   16
3        S002    D   17

School Code: S003
  School_Code Name  Age
4        S003    E   16

Type of GroupBy Object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>
```

EXP-17

```
import pandas as pd

df = pd.DataFrame({
    'School_Code':['S001','S002','S001','S002','S003'],
    'Age':[15,16,14,17,18]
})

result = df.groupby('School_Code')['Age'].agg(['mean','min','max'])

print(result)
```

```
             mean  min  max
School_Code
S001         14.5   14   15
S002         16.5   16   17
S003         18.0   18   18
```

EXP-18

```
import pandas as pd

df = pd.DataFrame({
    'School_Code':['S001','S001','S002','S002','S003'],
    'Class':['V','VI','V','VI','V'],
    'Name':['A','B','C','D','E']
})

grouped = df.groupby(['School_Code','Class'])

for name, group in grouped:
    print("\nGroup:", name)
    print(group)
```

```
...
Group: ('S001', 'V')
  School_Code Class Name
0        S001     V    A

Group: ('S001', 'VI')
  School_Code Class Name
1        S001    VI    B

Group: ('S002', 'V')
  School_Code Class Name
2        S002     V    C

Group: ('S002', 'VI')
  School_Code Class Name
3        S002    VI    D

Group: ('S003', 'V')
  School_Code Class Name
4        S003     V    E
```

EXP-19

```
import pandas as pd

data = {
    'country': ['India', 'USA', 'UK'],
    'year': [2015, 2015, 2015],
    'spirit_servings': [5, 100, 80],
    'wine_servings': [1, 20, 50],
    'beer_servings': [10, 200, 150]
}

df = pd.DataFrame(data)

print("Shape of Dataset:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())
```

```
Shape of Dataset:
(3, 5)

Column Names:
['country', 'year', 'spirit_servings', 'wine_servings', 'beer_servings']
```

EXP-20

```
import pandas as pd

df = pd.DataFrame({
    'Name':['John', 'Alice', 'Johnson', 'Bob']
})

result = df['Name'].str.find('son')

print(result)
```

```
... 0   -1
1   -1
2    4
3   -1
Name: Name, dtype: int64
```